

Safwen Cherif

Software Engineering Student — Co-Founder @ 3aweni.tn

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Professional Summary

Software Engineering student specializing in AI, Data Engineering, and scalable software systems, with hands-on experience building full-stack applications and end-to-end MLOps pipelines. Passionate about MLOps, DataOps, and the application of AI in healthcare. Co-founder of **3aweni.tn**, a startup incubated at GFI, where I lead product architecture and software development from concept to deployment.

Education

Ecole Polytechnique Sousse

2024 – Present

Software Engineering and Information Systems – Engineering Degree (Diplôme d'Ingénieur)

Higher Institute of Information and Communication Technologies

2021 – 2024

Bachelor's Degree in Software Engineering and Information Systems

Entrepreneurial Experience

3aweni.tn (Startup, Incubated by GFI)

2025 – Present

Co-Founder & Software Engineer

- Leading design and development of a multi-platform digital services product from concept to nearing production release on Android, now expanding to iOS and Web.
- Collaborating with the GFI Incubator on legal structuring and go-to-market preparation, including the Tunisian Startup Act application.
- Stack: **Flutter, Next.js, NestJS, Python, Firebase, Supabase, LLM APIs.**

Professional Experience – Internships

Elite Council Consulting

Jun – Jul 2025

Software Development Intern (Remote) – React.js, Flask, MongoDB

- Built a smart school attendance system with AI integration, RESTful APIs, and real-time monitoring dashboards.
- Implemented authentication, role-based access control, and data visualization features.

Nachd-IT

Feb 2024 – May 2024

Final Year Project Intern (Sousse) – Laravel, JavaScript, jQuery

- Designed and developed **Hannibal Market**, a multi-vendor e-commerce platform.
- Integrated real-time messaging and vendor management features.
- Followed agile practices and delivered a complete production-ready solution.

Nachd-IT

Jul 2023 – Aug 2023

Software Development Intern (Sousse) – Laravel, MySQL

- Developed **MyLabo**, a dynamic website for a research laboratory.
- Implemented backend logic, database design, and responsive UI components.

Key Projects

Hybrid Anime Recommendation System – End-to-End MLOps Pipeline

- Built and deployed a production-grade hybrid recommendation engine combining **TensorFlow/Keras embedding-based collaborative filtering** with content-based similarity for personalized top-10 recommendations.
- Designed a complete MLOps workflow with **DVC data versioning on Google Cloud Storage, Comet ML experiment tracking**, Docker containerization, and automated Jenkins CI/CD deployment.
- Deployed the Flask recommendation service on **Google Kubernetes Engine (GKE)** using Kubernetes Deployment and LoadBalancer for scalable cloud serving.
- Stack: **Python, TensorFlow, Keras, Flask, DVC, Comet ML, Docker, Jenkins, Kubernetes, GCP.**

Hotel Reservation Cancellation Prediction – MLOps Project

- Developed an end-to-end ML system predicting hotel booking cancellations.
- Automated the complete ML lifecycle: data ingestion from **Google Cloud Storage**, preprocessing, SMOTE balancing, feature selection, model training, and experiment tracking with **MLflow**.
- Built a Flask prediction API containerized with Docker and deployed automatically to **Google Cloud Run** through Jenkins CI/CD pipelines.
- Stack: **Python, LightGBM, Scikit-learn, MLflow, Flask, Docker, Jenkins, GCP Cloud Run**.

MLOps Pipeline for Real Estate Data (Mubawab.tn)

- Built an end-to-end ML pipeline with web scraping, geographic data enrichment, EDA, modeling, and an automated retraining pipeline orchestrated with Apache Airflow.
- Stack: **Python, Geopy, MLflow, DVC, Airflow, FastAPI, Scikit-learn, Pandas**.

Food Delivery Microservices Platform

- Designed a microservices-based food delivery system (Catalogue, Order, Delivery) using gRPC and REST, with an event-driven architecture on Apache Kafka for real-time synchronization.
- Built an API Gateway exposing REST and GraphQL (Apollo Server) endpoints with cross-service data aggregation.
- Stack: **Node.js, Express, gRPC, GraphQL, Kafka, SQLite, RxDB**.

Data Warehouse Project – Power BI vs. Python

- Designed a Data Warehouse on the Northwind database with ETL pipelines and analytical dashboards using **Python, SQLAlchemy, Power BI**.

Other Projects

- **Mobile E-Commerce Application:** Android app with delivery tracking and in-app messaging (Android Studio, Java, Firebase, Cloudinary).
- **MyBloggy:** Full-stack blogging platform (React.js, Node.js, Express.js, MongoDB).
- **Hospital Management System:** Desktop app for managing patients, doctors, and appointments with authentication and CRUD operations (Java Swing).

Technical Skills

Languages: Python, Java, JavaScript, PHP, C, Dart

Frameworks: React.js, Next.js, Nest.js, Node.js, Laravel, Symfony, Flask, Django, Flutter, FastAPI

Data & AI: Pandas, NumPy, Scikit-learn, TensorFlow, Keras, LightGBM, MLflow, DVC, Comet ML, Airflow, ETL, Data Warehousing, Power BI

Databases: MySQL, MongoDB, SQLite, SQL Server, AWS S3 buckets

Tools: Git, Docker, Jenkins, Kubernetes, Postman, Firebase, Android Studio, UML

Cloud Tools: AWS, Azure, Google Cloud Platform

Practices: Agile, Scrum

Certifications & Professional Development

- IBM Big Data Engineer Certificate
- Python Specialist Certification (Certiport)
- Supervised & Unsupervised Machine Learning – DataCamp
- MLOps — Machine Learning Operations Specialization – Duke University (Coursera) — In Progress
- AI for Medicine Specialization – DeepLearning.AI (Coursera) — In Progress

Languages

Arabic (Native) — English (Fluent) — French (Fluent)

Activities & Volunteering

- Red Crescent, Chebba (2022–Present) – Treasurer, "Koffet Romdhan" initiative
- ACM EPS, Sousse (2024–Present) – Member
- Polytech Sousse PSC (2024) – Public health initiative for primary schools with the Regional Health Delegation of Sousse